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import os
import sys
import time
import requests
import yfinance as yf
from curl_cffi import requests as cffi_requests
from datetime import datetime
import pytz

BOT_TOKEN = os.environ['TELEGRAM_BOT_TOKEN']
CHAT_ID   = os.environ['TELEGRAM_CHAT_ID']
TIMEZONE  = 'Asia/Bangkok'
TOP_N     = 3

WATCHLIST = [
    'AAPL', 'MSFT', 'GOOGL', 'AMZN', 'META', 'NVDA', 'BRK-B',
    'JPM', 'V', 'MA', 'UNH', 'JNJ', 'PG', 'KO', 'WMT',
    'HD', 'COST', 'ADBE', 'CRM', 'NFLX', 'TSM', 'AVGO',
    'LLY', 'TMO', 'ABBV', 'MCD', 'NKE', 'SBUX', 'DIS',
    'SCHW', 'CI', 'TTD', 'FDS', 'KR',
]

def send(text):
    r = requests.post(
        'https://api.telegram.org/bot' + BOT_TOKEN + '/sendMessage',
        json={'chat_id': CHAT_ID, 'text': text, 'parse_mode': 'HTML'},
        timeout=15,
    )
    r.raise_for_status()
    time.sleep(0.5)

def fetch(ticker):
    try:
        session = cffi_requests.Session(impersonate='chrome')
        info = yf.Ticker(ticker, session=session).info
        price = info.get('currentPrice') or info.get('regularMarketPrice')
        if not price:
            return None
        return {
            'ticker':      ticker,
            'name':        info.get('shortName', ticker),
            'price':        float(price),
            'eps_fwd':      info.get('forwardEps'),
            'eps_ttm':      info.get('trailingEps'),
        }
    
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        'growth':      info.get('earningsGrowth') or info.get('revenueGrowth'),
        'rev_growth':  info.get('revenueGrowth'),
        'fcf':         info.get('freeCashflow'),
        'shares':      info.get('sharesOutstanding'),
        'roe':         info.get('returnOnEquity'),
        'net_margin':  info.get('profitMargins'),
        'gross_margin': info.get('grossMargins'),
        'peg':         info.get('pegRatio'),
        'insider_pct': info.get('heldPercentInsiders'),
        'desc':        (info.get('longBusinessSummary') or '')[:120],
    }
except Exception as e:
    print(ticker + ' error: ' + str(e))
    return None

def qscore(d):
    s = 0
    if d['roe'] and d['roe'] >= 0.15: s += 25
    if d['net_margin'] and d['net_margin'] >= 0.15: s += 25
    if d['gross_margin'] and d['gross_margin'] >= 0.40: s += 25
    if d['rev_growth'] and d['rev_growth'] >= 0.10: s += 25
    return s

def make_row(d, fair, buy, gap, metrics):
    return {
        'ticker': d['ticker'],
        'name':    d['name'],
        'score':   qscore(d),
        'status':  'BUY ZONE' if d['price'] <= buy else 'WATCH',
        'price':   d['price'],
        'fair':    fair,
        'buy':     buy,
        'disc':    (d['price'] - fair) / fair * 100,
        'metrics': metrics,
        'desc':    d['desc'],
        'gap':     gap,
    }

def buffett(d):
    eps = d['eps_fwd']
    if not eps or eps <= 0:
        return None
    fair = eps * 22
    buy  = fair * 0.85
    gap  = (d['price'] - buy) / buy
    if gap > 0.10:
        return None

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m = []
if d['roe']:          m.append('ROE ' + str(round(d['roe']*100)) + '%')
if d['net_margin']:   m.append('NM ' + str(round(d['net_margin']*100)) + '%')
if d['gross_margin']: m.append('GM ' + str(round(d['gross_margin']*100)) + '%')
return make_row(d, fair, buy, gap, ' / '.join(m))

def lynch(d):
    eps    = d['eps_ttm'] or d['eps_fwd']
    growth = d['growth']
    if not eps or eps <= 0 or not growth or growth <= 0:
        return None
    gp    = growth * 100
    fair  = eps * gp
    buy   = fair * 0.95
    gap   = (d['price'] - buy) / buy
    if gap > 0.10:
        return None
    peg = ('PEG ' + str(round(d['peg'], 2))) if d['peg'] else ('PEG ~' + str(round(d['peg'], 2)))
    m    = [peg, 'EPS +' + str(round(gp)) + '%']
    if d['rev_growth']: m.append('Rev +' + str(round(d['rev_growth']*100)) + '%')
    return make_row(d, fair, buy, gap, ' / '.join(m))

def sleep_scan(d):
    if not d['fcf'] or not d['shares'] or d['shares'] <= 0:
        return None
    fcf_ps = d['fcf'] / d['shares']
    if fcf_ps <= 0:
        return None
    fair = fcf_ps / 0.04
    buy  = fair * 0.90
    gap  = (d['price'] - buy) / buy
    if gap > 0.10:
        return None
    m = []
    if d['rev_growth']: m.append('Rev +' + str(round(d['rev_growth']*100)) + '%')
    if d['roe']:        m.append('ROE ' + str(round(d['roe']*100)) + '%')
    if d['insider_pct']: m.append('Insider ' + str(round(d['insider_pct']*100, 1)) + '%')
    return make_row(d, fair, buy, gap, ' / '.join(m))

def fmt(rank, s, prefix=''):
    emoji = '🟢' if 'BUY' in s['status'] else '🟡'
    lines = [
        '',
        '# ' + str(rank) + ' ' + prefix + s['ticker'] + ' - ' + s['name'],
        '  Quality: ' + str(s['score']) + '/100 | ' + emoji + ' ' + s['status'],
        '  Price: $' + str(round(s['price'], 2)) + ', ',
        '  Fair: $' + str(round(s['fair'], 2)) + ', '
    ]

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        ' (' + str(round(s['disc'])) + '%)' +
        ' | Buy: $' + str(round(s['buy'], 2)),
        ' ' + s['metrics'],
    ]
    if s['desc']:
        lines.append(' ' + s['desc'])
    return '\n'.join(lines)

def main():
    print('Scanning ' + str(len(WATCHLIST)) + ' tickers...')
    buf, lyn, slp = [], [], []

    for ticker in WATCHLIST:
        d = fetch(ticker)
        if not d:
            continue
        r = buffett(d)
        if r: buf.append(r)
        r = lynch(d)
        if r: lyn.append(r)
        r = sleep_scan(d)
        if r: slp.append(r)

    def sort_key(x):
        return (x['gap'], -x['score'])

    buf.sort(key=sort_key)
    lyn.sort(key=sort_key)
    slp.sort(key=sort_key)

    today = datetime.now(pytz.timezone(TIMEZONE)).strftime('%d %b %Y')

    send(
        '<b>VI Daily Scan - ' + today + '</b>\n'
        'BUY ZONE = price below buy target\n'
        'WATCH = within 10% above buy target\n'
        'Always verify before investing.'
    )

    if buf:
        body = ''.join(fmt(i+1, s) for i, s in enumerate(buf[:TOP_N]))
        send('<b>WARREN BUFFETT Style</b>\n(Fair=EPS x 22 / Buy=Fair x 0.85)' + b

    if lyn:
        body = ''.join(fmt(i+1, s, 'Fast Grower: ') for i, s in enumerate(lyn[:TO
        send('<b>PETER LYNCH Style</b>\n(Fair=EPS x Growth / Buy=Fair x 0.95)' +

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if slp:
    body = ''.join(fmt(i+1, s) for i, s in enumerate(slp[:TOP_N]))
    send('<b>NICK SLEEP Style</b>\n(Fair=FCF/0.04 / Buy=Fair x 0.90)' + body)

if not buf and not lyn and not slp:
    send('VI Daily Scan - No stocks in BUY/WATCH zone today.')

print('Done.')

if __name__ == '__main__':
    main()
```